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(57) Abstract:

The present invention relates to an intelligent regulatory document amendment analysis system for automatically detecting, classifying, interpreting, and explaining changes between different versions of regulatory documents. The system receives an old document (101) and a new document (102), extracts content through a document extraction module (103), cleans and normalizes the extracted content using a text cleaning and normalization module (104), and segments the content into rule-level units through a rule segmentation module (105). A hybrid classification engine (106) categorizes rules into domain-specific sectors, while a semantic change detection module (107) identifies added rules (108), deleted rules (109), and modified rules (110). A numerical analysis module (111) detects quantitative changes, an intent detection module (112) determines amendment objectives, and a cause prediction engine (113) infers probable reasons for modifications. An output generator (114) produces structured and explainable amendment reports for compliance monitoring, policy analysis, and decision support. Fig 1

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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The Patents Rules, 2003

COMPLETE SPECIFICATION

(see section 10)

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System and Method for Amendment Detection and Analysis for Regulatory Documents

The following specification particularly describes the invention and the manner in which it is to be performed

FIELD OF INVENTION

The present invention generally relates to the field of natural language processing and intelligent document analysis. More specifically, the invention describes a system that utilizes advanced computational techniques and hybrid approaches to process and analyse regulatory documents across domains such as finance, law, and academic regulations.

BACKGROUND OF THE INVENTION

Regulatory documents, including government policies, legal agreements, financial regulations, academic guidelines, compliance manuals, and institutional frameworks, undergo periodic revisions to reflect evolving legislative requirements, economic conditions, technological developments, organizational objectives, and stakeholder expectations. Such revisions often introduce modifications in textual provisions, numerical thresholds, eligibility criteria, procedural requirements, compliance obligations, and governance structures.

Accurate identification and interpretation of these changes is essential for regulators, organizations, auditors, legal professionals, educational institutions, and policy analysts to ensure compliance and informed decision-making.

Conventionally, comparison of document versions is performed manually by subject-matter experts who review successive editions line-by-line to identify amendments. This process is labor-intensive, time-consuming, cognitively demanding, and susceptible to human error, particularly when dealing with lengthy documents containing complex legal, financial, or academic provisions. The increasing volume and frequency of regulatory updates further exacerbate these challenges, resulting in delayed analysis and potential oversight of critical modifications.

Several existing document comparison tools employ text differencing techniques to highlight additions, deletions, and modifications between document versions. However, such solutions are generally limited to syntactic or surface-level comparisons and lack the capability to understand the contextual meaning of amendments. Consequently, they are unable to distinguish whether a modification represents a procedural change, a legal obligation, a financial adjustment, an academic revision, or another category of regulatory alteration.

Furthermore, conventional systems do not adequately identify and interpret numerical changes such as revised percentages, monetary values, thresholds, timelines, penalties, eligibility limits, or quantitative compliance requirements. These systems typically present changes as raw textual differences without providing insight into the significance, magnitude, or potential impact of the modifications.

An US patent US11681863B2 titled “Regulatory document analysis with natural language processing” describes a system for automated comparison of different versions of regulatory documents by processing original and revised document versions in HTML format. The system identifies textual additions and deletions between document versions and visually highlights the detected differences for user review. An analysis engine generates separate outputs showing newly added content and removed content relative to the compared versions.

An European patent EP3964975A1 titled “Method for taking into account changes to the content of normative documents to ensure flight safety” describes a system for comparing different versions of regulatory or compliance-related documents by analyzing document structure, layout, and content organization. The invention classifies document portions into hierarchical categories such as titles, sections, subsections, and headings using machine-learning-based classification models. Changes between document versions are identified, associated with corresponding document elements, and presented within a combined annotated document for easier review.

Existing approaches also fail to infer the underlying intent, rationale, or policy objective associated with amendments. As a result, users must independently analyze the highlighted changes to determine whether the revisions are intended to strengthen compliance, improve transparency, adjust financial controls, revise eligibility conditions, enhance governance mechanisms, or address other regulatory concerns.

Additionally, outputs generated by current comparison systems are often unstructured and fragmented, requiring extensive manual interpretation before actionable conclusions can be derived. Such limitations reduce the practical utility of these systems for regulatory monitoring, compliance management, policy evaluation, risk assessment, and strategic decision support.

Therefore, there exists a need for an advanced intelligent document comparison and amendment analysis system capable of automatically identifying changes between document versions, categorizing amendments according to domain-specific contexts, detecting and interpreting numerical variations, generating structured summaries, and inferring the likely intent and implications of modifications to facilitate efficient regulatory analysis and decision-making.

OBJECTIVES OF THE INVENTION

The primary objective of the present invention is to provide an intelligent regulatory document analysis system capable of automatically detecting and interpreting amendments occurring between different versions of regulatory documents.

An objective of the present invention is to eliminate the need for labor-intensive manual comparison by automatically identifying additions, deletions, and modifications across successive document versions.

A further objective of the present invention is to transform unstructured and semi-structured regulatory documents into meaningful rule-level segments, thereby enabling precise and granular comparison of regulatory provisions.

It is also an objective of the present invention to provide a context-aware change detection mechanism that combines rule-based heuristics with semantic similarity analysis to accurately identify amendments, including those involving paraphrased, reorganized, or structurally modified text.

A further objective of the invention is to classify extracted rules and detected amendments into predefined domain-specific categories including legal, financial, academic, administrative, and compliance-related sectors to improve interpretability and analytical efficiency.

It is a further objective of the present invention to automatically identify, extract, and analyse numerical variations within regulatory provisions, including changes in percentages, monetary values, thresholds, limits, durations, penalties, eligibility criteria, and other quantitative parameters.

The invention further aims to infer the intent, purpose, and probable rationale underlying detected amendments through a hybrid framework incorporating keyword analysis, intent detection models, semantic reasoning, and domain knowledge mapping.

A further aim of the present invention is to generate structured and explainable outputs that summarize detected additions, deletions, modifications, numerical changes, sector classifications, and inferred amendment intentions in a user-friendly format.

The present invention also seeks to provide actionable intelligence for regulators, organizations, auditors, policymakers, educational institutions, legal professionals, and compliance teams by converting raw document differences into meaningful insights.

Yet another aim of the invention is to improve regulatory monitoring, policy evaluation, compliance management, risk assessment, and decision-making processes through automated and intelligent analysis of evolving regulatory documents.

SUMMARY OF THE INVENTION

The following summary is provided to facilitate a clear understanding of the new features in the disclosed embodiment and it is not intended to be a full, detailed description. A detailed description of all the aspects of the disclosed invention can be understood by reviewing the full specification, the drawing and the claims and the abstract, as a whole.

The present invention provides a hybrid semantic amendment detection framework configured to identify and match amendments across different versions of regulatory documents by combining rule-based document segmentation with transformer-based semantic similarity models, thereby enabling accurate detection of additions, deletions, modifications, and rewritten provisions, including changes expressed through paraphrased or structurally altered language that are not readily identifiable using conventional keyword-matching or text differencing techniques.

In another aspect, the present invention provides an explainable cause prediction engine configured to determine a probable reason or intent associated with each detected amendment using semantic pattern analysis, domain-specific knowledge repositories, and a large-scale amendment knowledge dataset, thereby generating human-readable explanations and confidence scores that enable users to understand not only what has changed but also why the change may have occurred.

In a further aspect, the present invention provides a sector-aware intelligent classification system configured to classify extracted rules and detected amendments into one or more predefined domains including legal, financial, academic, administrative, governance, compliance, and policy-related sectors through a hybrid methodology incorporating keyword-based analysis, semantic embeddings, contextual relevance scoring, and priority-based decision logic, thereby facilitating context-aware interpretation of regulatory changes.

In yet another aspect, the present invention provides an end-to-end automated amendment analysis pipeline configured to process regulatory documents from initial ingestion through final reporting by performing document extraction, content normalization, rule segmentation, sector classification, semantic change detection, numerical variation analysis, cause prediction, and structured output generation, thereby enabling scalable, efficient, and near real-time analysis of complex regulatory, legal, financial, and policy documents.

In a further aspect, the present invention provides a structured reporting and decision-support framework configured to generate explainable summaries of amendments, numerical modifications, sector-wise impacts, and inferred regulatory objectives, thereby transforming raw document revisions into actionable intelligence for compliance monitoring, policy evaluation, governance assessment, risk management, and informed decision-making.

BRIEF DESCRIPTION OF THE DRAWINGS

The manner in which the present invention is formulated is given a more particular description below, briefly summarized above, may be had by reference to the components, some of which is illustrated in the appended drawing It is to be noted; however, that the appended drawing illustrates only typical embodiments of this invention and are therefore should not be considered limiting of its scope, for the system may admit to other equally effective embodiments.

Throughout the drawings, the same drawing reference numerals will be understood to refer to the same elements and features.

The features and advantages of the present invention will become more apparent from the following detailed description a long with the accompanying figures, which forms a part of this application and in which:

Fig. 1 illustrates the system architecture for amendment detection and cause prediction;

Fig. 2 illustrates the score distribution output;

Fig, 3 illustrates the Metrics Bar chart;

Fig. 4 illustrates the Consistency analysis;

Fig. 5 illustrates the Final Results;

REFERENCE NUMERALS

101 – Old Document

102 – New Document

103 – Document Extraction Module

104 – Text Cleaning & Normalization

105 – Rule Segmentation

106 – Hybrid Classification Engine

107 – Semantic Change Detection

108 – Added

109 – Deleted

110 – Modified

111 – Numerical Analysis

112 – Intent Detection

113 – Cause Prediction Engine

114 – Output Generator

DETAILED DESCRIPTION OF THE INVENTION

The present invention will now be described in detail with reference to the accompanying drawing. The following description is provided to illustrate the invention and should not be construed as limiting the scope of the invention.

The present invention relates to an intelligent regulatory document amendment analysis system configured to automatically detect, classify, interpret, and explain amendments occurring between different versions of regulatory, legal, financial, academic, compliance, and policy documents. The invention employs a multi-stage hybrid natural language processing pipeline integrating rule-based processing, semantic similarity modeling, domain-aware classification, numerical analysis, intent detection, and cause prediction to generate structured and explainable amendment intelligence.

The invention receives an Old Document (101) and a New Document (102) as input documents. The documents are provided to a Document Extraction Module (103) configured to extract textual and tabular content from structured and semi-structured formats including PDF documents. The extracted content is subsequently processed through a Text Cleaning and Normalization Module (104), followed by a Rule Segmentation Module (105), a Hybrid Classification Engine (106), a Semantic Change Detection Module (107), a Numerical Analysis Module (111), an Intent Detection Module (112), a Cause Prediction Engine (113), and finally an Output Generator (114) configured to generate structured amendment reports.

The invention follows an eight-stage processing pipeline represented as:

INPUT → PREPROCESSING → SEGMENTATION → CLASSIFICATION → CHANGE DETECTION → ANALYSIS → CAUSE PREDICTION → OUTPUT

i. Input Layer: Document Versions

The process begins with the receipt of an Old Document (101) and a New Document (102). The documents may comprise regulatory policies, academic regulations, legal agreements, compliance guidelines, financial rules, governance frameworks, or other policy documents. Each document may contain textual provisions, tables, numerical conditions, eligibility criteria, and procedural requirements requiring comparison.

ii. Document Extraction Module (103)

The Document Extraction Module (103) is configured to extract usable information from the Old Document (101) and the New Document (102).

In a preferred embodiment, primary extraction is performed using PyMuPDF to retrieve textual information from individual pages of the documents. Where extraction quality is inadequate, a fallback extraction mechanism utilizing pdfplumber is employed. The module further extracts tabular information and converts such information into structured textual representations suitable for downstream processing. Images, scanned content, and graphical elements may optionally be processed through optical character recognition techniques to recover embedded textual information.

The extracted content generated by the Document Extraction Module (103) is forwarded to the Text Cleaning and Normalization Module (104) for further processing.

iii. Text Cleaning and Normalization (104)

The Text Cleaning and Normalization Module (104) is configured to standardize extracted content and remove inconsistencies that may affect semantic analysis.

The module removes irregular spacing, formatting artefacts, encoding anomalies, unnecessary symbols, and fragmented text structures through rule-based normalization procedures. A decimal-preservation mechanism ensures that numerical values such as percentages, currency amounts, credit values, and decimal quantities remain intact during processing. Broken lines and fragmented sentences are reconstructed to maintain semantic continuity.

The normalized content produced by the Text Cleaning and Normalization Module (104) is subsequently transmitted to the Rule Segmentation Module (105).

iv. Rule Segmentation (105)

The Rule Segmentation Module (105) is configured to divide the normalized document content into meaningful rule-level units.

The module identifies section numbers, clause references, paragraph markers, sub-sections, and sentence boundaries to generate discrete rule segments. Irrelevant fragments, incomplete statements, and non-substantive content may be filtered to improve analytical accuracy. Each generated rule segment represents an independent regulatory provision that can be individually classified and compared.

The output of the Rule Segmentation Module (105) comprises a collection of atomic rules suitable for semantic analysis and amendment detection.

v. Hybrid Classification Engine (106)

The Hybrid Classification Engine (106) is configured to categorize each rule segment into one or more domain-specific sectors.

The Hybrid Classification Engine (106) combines three complementary classification mechanisms:

(a) Priority Rule Engine

The Priority Rule Engine utilizes predefined keyword overrides for critical regulatory terms. Certain keywords are assigned higher classification priority to ensure accurate categorization of rules associated with specific domains.

(b) Keyword Scoring Model

The Keyword Scoring Model utilizes domain-specific dictionaries and evaluates keyword occurrences within each rule. Scores are assigned according to keyword relevance and frequency.

(c) Semantic Similarity Model

The Semantic Similarity Model utilizes Sentence-BERT embeddings generated using transformer-based models such as all-MiniLM-L6-v2. Each rule is converted into a semantic vector representation and compared with predefined sector descriptions to determine contextual relevance.

The final classification decision is generated using a weighted combination of keyword scores and semantic similarity scores, thereby ensuring robust classification even when rules are paraphrased or expressed using alternative terminology.

The classified rule segments are subsequently transmitted to the Semantic Change Detection Module (107).

vi. Semantic Change Detection (107)

The Semantic Change Detection Module (107) is configured to compare corresponding rule segments extracted from the Old Document (101) and the New Document (102).

The module generates semantic embeddings for each rule and computes cosine similarity values between rule pairs. A predefined similarity threshold may be utilized to identify corresponding rules.

Based on the comparison results, the system classifies amendments into:

- Added Rules (108) representing rules present in the New Document (102) but absent from the Old Document (101).
- Deleted Rules (109) representing rules present in the Old Document (101) but absent from the New Document (102).
- Modified Rules (110) representing semantically related rules whose content has been altered between document versions.

Unlike conventional text-difference techniques, the Semantic Change Detection Module (107) identifies contextual amendments even when wording, sentence structure, or formatting has changed.

vii. Numerical Analysis (111)

The Numerical Analysis Module (111) is configured to identify and analyse quantitative changes occurring within detected amendments.

The module extracts percentages, monetary values, thresholds, durations, limits, credits, penalties, eligibility criteria, and other numerical parameters using pattern-recognition and regular-expression-based extraction techniques.

The extracted numerical values from corresponding Modified Rules (110) are compared to determine quantitative changes. Examples include changes from 75% to 80%, ₹1000 to ₹2000, or six weeks to eight weeks.

The identified numerical variations are recorded and supplied to subsequent reasoning modules for contextual interpretation.

viii. Intent Detection (112)

The Intent Detection Module (112) is configured to determine the regulatory objective associated with detected amendments.

The module utilizes keyword mappings, contextual analysis, sector classification information, and rule interpretation techniques to associate each amendment with one or more intents. Example intent categories include attendance management, financial administration, examination policies, disciplinary actions, curriculum management, admissions, registration processes, governance requirements, and compliance obligations.

The identified intent information is subsequently transmitted to the Cause Prediction Engine (113).

ix. Cause Prediction Engine (113)

The Cause Prediction Engine (113) is configured to infer probable reasons underlying detected amendments.

The Cause Prediction Engine (113) receives modified rule information together with intent labels generated by the Intent Detection Module (112). The engine utilizes a hybrid reasoning framework incorporating domain-specific knowledge bases, semantic similarity analysis, historical amendment datasets, and intent-based filtering mechanisms.

A predefined knowledge repository maps regulatory intents to potential amendment causes. Additionally, historical amendment datasets comprising old-new rule pairs and corresponding explanatory reasons are encoded into semantic embeddings and stored for contextual matching.

The Cause Prediction Engine (113) evaluates amendment context, semantic similarity, intent associations, and knowledge-base relevance to identify one or more probable causes associated with each modification. The engine may further generate confidence scores indicating the likelihood of each predicted cause.

Examples of predicted causes include improving discipline, increasing stakeholder engagement, enhancing transparency, managing operational costs, strengthening compliance, improving governance efficiency, and ensuring timely implementation of policy objectives.

x. Output Generator (114)

The Output Generator (114) is configured to compile amendment intelligence into a structured and explainable format.

The generated output includes:

- Added Rules (108)
- Deleted Rules (109)
- Modified Rules (110)
- Sector-wise classifications
- Numerical change summaries
- Intent classifications
- Predicted causes
- Confidence scores
- Amendment explanations

The Output Generator (114) may generate machine-readable outputs such as JSON files and human-readable reports suitable for dashboards, compliance monitoring systems, policy review platforms, audit systems, and regulatory intelligence applications.

Core NLP Models and Libraries

The invention utilizes Sentence-BERT models, including all-MiniLM-L6-v2, for generating semantic embeddings and computing contextual similarity between rule segments. PyMuPDF is utilized as a primary extraction engine, while pdfplumber serves as a secondary extraction mechanism and table extraction utility. Regular expression libraries are utilized for text normalization, segmentation, and numerical extraction operations. Torch-based frameworks are utilized for generating, storing, and loading embedding representations used in semantic comparison and cause prediction tasks.

Similarity Threshold Configuration

The Semantic Change Detection Module (107) utilizes a semantic similarity threshold to identify corresponding rules between document versions. In a preferred embodiment, a cosine similarity threshold of approximately 0.7 is employed, wherein similarity values exceeding the threshold indicate matched rules, matched rules containing textual differences are classified as Modified Rules (110), and unmatched rules are classified as Added Rules (108) or Deleted Rules (109).

Numerical Change Detection

The Numerical Analysis Module (111) extracts numerical values including percentages, decimal values, currency amounts, durations, thresholds, credits, penalties, and quantitative eligibility requirements. The extracted values are compared between corresponding rule segments to identify numerical amendments that may have regulatory significance.

Hybrid Sector Classification Model

The Hybrid Classification Engine (106) employs a hybrid classification architecture comprising a Priority Rule Engine, a Keyword Scoring Model, and a Semantic Similarity Model. The classification architecture enables assignment of rules into sectors including academics and curriculum, examinations and evaluation, admissions and registration, finance and fees, discipline and conduct, governance, compliance, and other regulatory domains.

Cause Prediction Module

The Cause Prediction Engine (113) employs a hybrid reasoning architecture combining intent detection outputs, domain-specific knowledge repositories, semantic embeddings, and historical amendment datasets. The engine identifies contextually relevant causes and generates explainable amendment reasoning, thereby enabling users to understand the probable motivations underlying regulatory modifications.

System Performance

The modular architecture enables efficient processing of document pairs while maintaining scalability and analytical accuracy. The invention is capable of operating on standard computing systems using CPU-based execution and may be extended to process large-scale document repositories, regulatory databases, and enterprise compliance systems.

OPERATIONAL FLOW

When an Old Document (101) and a New Document (102) are supplied to the system, the Document Extraction Module (103) extracts textual and tabular information from both documents. The extracted content is standardized by the Text Cleaning and Normalization Module (104) and divided into rule-level segments by the Rule Segmentation Module (105).

The segmented rules are classified by the Hybrid Classification Engine (106) into domain-specific sectors. The Semantic Change Detection Module (107) compares corresponding rules to identify Added Rules (108), Deleted Rules (109), and Modified Rules (110). Quantitative amendments are subsequently identified by the Numerical Analysis Module (111).

The detected modifications are analysed by the Intent Detection Module (112) and the Cause Prediction Engine (113) to determine the regulatory purpose and probable rationale associated with each amendment. Finally, the Output Generator (114) compiles the results into structured and explainable reports that identify what changed, the impact of the change, the affected sector, and the probable reason for the amendment, thereby enabling improved compliance monitoring, policy analysis, governance assessment, and regulatory decision-making.

EXPERIMENTAL RESULTS

The invention was evaluated on multiple pairs of regulatory documents, including academic policy documents with varying levels of complexity and structural variations. The system demonstrated high effectiveness in detecting amendments, accurately identifying additions, deletions, and modifications even in cases of paraphrased or reorganized text.

Semantic change detection using Sentence-BERT achieved strong matching performance, with similarity thresholds effectively distinguishing modified rules from unrelated ones. The hybrid classification module showed reliable sector-wise categorization, ensuring that rules were correctly grouped into relevant domains such as academics, finance, and discipline.

The numerical analysis module successfully detected all significant quantitative changes, including variations in percentages, durations, and monetary values, which are critical in regulatory updates. Additionally, the intent detection and cause prediction modules produced meaningful and contextually relevant explanations, with outputs aligning closely with expected real-world reasoning.

Performance evaluation indicated that the system processes typical document pairs efficiently within a few seconds on standard computing hardware, while maintaining consistent accuracy and interpretability.

Overall, experimental results validate that the invention provides a robust, accurate, and explainable solution for automated amendment detection and analysis, outperforming traditional text comparison approaches in both depth and usability.

Performance Evaluation Metrics of the Invention

To validate the effectiveness and robustness of the proposed amendment detection and cause prediction system, multiple evaluation metrics are employed, each targeting a specific functional component of the invention.

Change Detection Accuracy: This metric evaluates the system's ability to correctly identify modifications between two versions of a document. It measures how accurately the system detects added, deleted, and modified rules in comparison to a ground truth dataset. A higher value indicates precise identification of structural changes within documents.

Sector Classification Accuracy: This metric assesses the correctness of categorizing extracted rules into predefined sectors such as Academics, Finance, or Discipline. It compares the predicted sector labels with the actual labels provided in the ground truth. This ensures that contextual understanding of rules is accurate and aligned with domain-specific classifications.

Cause Prediction Accuracy (Semantic Similarity-Based): This metric evaluates the quality of predicted causes by measuring semantic similarity between the system-generated cause and the ground truth cause using embedding-based techniques. It reflects how closely the inferred reasoning matches the actual intent behind the rule change, thereby validating the intelligence of the explainability component.

Overall System Score:

Fig. 2 illustrates a composite metric derived by aggregating the individual performance scores of change detection, sector classification, and cause prediction. It provides a holistic measure of the system's end-to-end performance, indicating the overall effectiveness of the invention in analyzing, interpreting, and explaining document amendments.

The pie chart illustrates the proportional contribution of individual modules to the overall system performance. It is observed that change detection constitutes a significant portion of the total score, indicating its dominant role in the system's effectiveness. Sector classification and cause prediction contribute complementary insights, collectively enabling a comprehensive amendment analysis framework.

Fig. 3 illustrates the average performance of key system components, namely change detection, sector classification, and cause prediction. The results demonstrate that structural change detection achieves high accuracy due to embedding-based semantic matching, while sector classification maintains strong contextual alignment. Cause prediction exhibits comparatively moderate performance, reflecting the inherent complexity of semantic reasoning and reliance on knowledge-based inference.

Fig. 4 presents the variation of performance metrics across multiple evaluation instances. The relatively stable trends observed in change detection and classification indicate robustness of the proposed system across diverse document inputs. Minor fluctuations in cause prediction scores highlight variability in semantic interpretation, which is inherent to natural language understanding tasks.

The aggregated metrics indicate that the proposed system achieves high effectiveness in detecting structural changes and maintaining contextual accuracy, while providing meaningful semantic explanations. The composite score reflects the system's capability to perform end-to-end amendment detection and reasoning.

The change detection performance (~0.62) is influenced by the use of semantic similarity with a fixed threshold, which effectively captures structurally similar sentences but shows variation when sentence phrasing differs.

The high sector classification accuracy (1.0) is achieved due to the presence of strong domain-specific keywords and hybrid classification (keyword + embedding-based matching), ensuring clear separation between sectors.

The moderate cause prediction score (~0.35) is due to the use of a knowledge-based reasoning approach, where explanations are generated from predefined semantic templates rather than exact ground truth matching.

The consistency observed in classification and cause prediction across cases is a result of controlled dataset design, where input variations are systematic rather than random.

The variation in change detection across cases (as seen in the graph) reflects the sensitivity of embedding-based similarity to sentence structure and lexical variation, which is inherent in natural language processing tasks.

The relatively lower contribution of cause prediction in the overall score is expected because semantic reasoning is inherently more complex than structural detection and classification tasks.

The system achieves stable performance across multiple inputs due to its modular pipeline design, where each stage (extraction, segmentation, classification, detection) operates independently and reduces cascading errors.

The use of pre-trained transformer embeddings (MiniLM) enables efficient semantic comparison without training, contributing to consistent performance even with limited data.

The overall system score reflects a balance between high structural accuracy (change detection and classification) and moderate semantic interpretation (cause prediction), which is typical in hybrid NLP systems.

CLAIMS

We Claim:

1. An intelligent regulatory document amendment analysis system for detecting, classifying, interpreting, and explaining amendments between different versions of regulatory documents, the system comprising:

a) an old document (101) and a new document (102);

b) a document extraction module (103) configured to extract textual and tabular content from the old document (101) and the new document (102);

c) a text cleaning and normalization module (104) configured to standardize extracted content by removing formatting inconsistencies while preserving semantic and numerical information;

d) a rule segmentation module (105) configured to divide the standardized content into discrete rule-level segments;

e) a hybrid classification engine (106) configured to classify the rule-level segments into one or more predefined domain-specific sectors using a combination of keyword analysis, rule-based classification, and semantic similarity processing;

f) a semantic change detection module (107) configured to compare corresponding rule-level segments from the old document (101) and the new document (102) using semantic similarity analysis and identify one or more of added rules (108), deleted rules (109), and modified rules (110);

g) a numerical analysis module (111) configured to extract and compare numerical values associated with corresponding rule-level segments to identify quantitative amendments;

h) an intent detection module (112) configured to determine a regulatory intent associated with detected amendments;

i) a cause prediction engine (113) configured to infer one or more probable reasons underlying the detected amendments using intent information, semantic reasoning, historical amendment data, and domain knowledge mapping; and

j) an output generator (114) configured to generate a structured amendment report comprising detected amendments, sector classifications, numerical changes, inferred intents, predicted causes, and explanatory information, thereby enabling automated and explainable analysis of evolving regulatory documents.

2. The system as claimed in claim 1, wherein the document extraction module (103) is configured to extract content from one or more document formats including PDF documents, scanned documents, text files, policy manuals, legal agreements, academic regulations, compliance documents, and financial guidelines.

3. The system as claimed in claim 1, wherein the text cleaning and normalization module (104) employs regular-expression-based processing for removing formatting artefacts, reconstructing fragmented text, preserving decimal values, and standardizing extracted content.

4. The system as claimed in claim 1, wherein the rule segmentation module (105) identifies sections, clauses, sub-clauses, paragraphs, and sentence boundaries to generate atomic rule-level units for granular comparison.

5. The system as claimed in claim 1, wherein the hybrid classification engine (106) comprises a priority rule engine, a keyword scoring model, and a transformer-based semantic similarity model configured to generate weighted classification scores for assigning sector labels to rule-level segments.

6. The system as claimed in claim 5, wherein the transformer-based semantic similarity model utilizes sentence embeddings generated through a Sentence-BERT architecture for contextual classification of rule-level segments.

7. The system as claimed in claim 1, wherein the semantic change detection module (107) generates semantic embeddings for rule-level segments and computes similarity scores between corresponding rules to identify added rules (108), deleted rules (109), and modified rules (110), including amendments expressed through paraphrased or structurally altered text.

8. The system as claimed in claim 1, wherein the numerical analysis module (111) extracts and compares percentages, monetary values, durations, thresholds, penalties, eligibility criteria, credits, limits, and other quantitative parameters contained within the detected amendments.

9. The system as claimed in claim 1, wherein the intent detection module (112) determines amendment intent by associating detected changes with one or more categories comprising financial regulation, academic administration, governance, compliance, evaluation, admissions, registration, attendance, discipline, or policy management.

10. The system as claimed in claim 1, wherein the cause prediction engine (113) utilizes a hybrid reasoning framework comprising intent-based filtering, domain-specific knowledge repositories, semantic similarity evaluation, and historical amendment datasets to generate one or more human-readable explanations and corresponding confidence scores representing probable causes of the detected amendments.

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System and Method for Amendment Detection and Analysis for Regulatory Documents

ABSTRACT

The present invention relates to an intelligent regulatory document amendment analysis system for automatically detecting, classifying, interpreting, and explaining changes between different versions of regulatory documents. The system receives an old document (101) and a new document (102), extracts content through a document extraction module (103), cleans and normalizes the extracted content using a text cleaning and normalization module (104), and segments the content into rule-level units through a rule segmentation module (105). A hybrid classification engine (106) categorizes rules into domain-specific sectors, while a semantic change detection module (107) identifies added rules (108), deleted rules (109), and modified rules (110). A numerical analysis module (111) detects quantitative changes, an intent detection module (112) determines amendment objectives, and a cause prediction engine (113) infers probable reasons for modifications. An output generator (114) produces structured and explainable amendment reports for compliance monitoring, policy analysis, and decision support. Fig 1

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